

Cheap Deterministic Shortcuts vs. LLM-as-a-Judge Metrics: An Agreement Study on a Real Agentic-RAG Memory Store

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Abstract

LLM-as-a-judge is the default method for evaluating retrieval-augmented generation (RAG) systems, but it is slow, non-deterministic, and expensive. We measure how well cheap, deterministic shortcuts—built from embeddings, cross-encoders, natural language inference, and lexical overlap—predict the rankings of four RAGAS judge metrics on a production-shaped corpus: the full session store of an agentic coding tool mirrored into a local vector database. Over 99 joined samples, AnswerCorrectness is strongly predicted by a ROUGE-L plus embedding-cosine blend (Spearman $\rho = 0.89$), and AnswerRelevancy by query/document cosine using the index’s own embedding family ($\rho = 0.71$). Faithfulness agreement is weakly negative ($\rho = -0.18$): the judge saturates at a median of 1.000 while a sentence-level NLI shortcut reads a median of 0.000. ContextPrecision shows no discriminative signal on this corpus ($|\rho| < 0.13$ across all variants). We conclude that shortcut-versus-judge agreement is a fast, actionable diagnostic: it identifies which judge metrics can be replaced by deterministic scores for ranking, and which judge metrics themselves require an oracle audit.

1 Introduction

Retrieval-augmented generation (RAG) is evaluated predominantly through LLM-as-a-judge frameworks such as RAGAS [1]. These metrics require a capable judge model, are non-deterministic, and take hours to run on realistic datasets with local models. Teams that iterate on retrieval pipelines need cheaper, deterministic signals for regression gates and system tuning.

In this work we ask an operational question: *on a given corpus, which RAGAS judge metric can be predicted by which cheap shortcut, and which judge metrics are themselves unreliable?* Rather than a synthetic benchmark, we evaluate on the session history of an agentic coding assistant—thousands of real conversations—mirrored into a local vector database. Queries are genuine user messages, responses are fresh generations conditioned on retrieved chunks, and references are the historical assistant replies.

2 System under evaluation

The retrieval system is a derived vector index of a SQLite session store. Messages are extracted from materialized tables into a JSONL corpus, chunked to roughly 1400 characters with overlap, embedded with `nomic-embed-text` (768-dimensional, cosine), and stored in a Chroma collection. Retrieval is top- k cosine search ($k = 5$); a local generator (`gemma4:12b`, temperature 0) answers given the retrieved chunks. The same model family serves as the RAGAS judge.

3 Method

3.1 Dataset

We sampled 100 query/response/context triples deterministically (seed 42, at most 5 per session) from the session store. Queries are user messages of 15–1500 characters; references are the historical assistant replies; retrieved contexts are the top-5 chunks for the query; responses are fresh generations conditioned on those chunks. After filtering invalid judge outputs, **99** samples were joined across all analyses.

3.2 Reference (judge) metrics

We use the classic RAGAS metric classes [1]: Faithfulness, AnswerRelevancy, ContextPrecision, and AnswerCorrectness, run with a fully local judge (**gemma4:12b** via Ollama) and **nomic-embed-text** embeddings, single worker, on the GPU. The full pass took 8 h 26 min for the 400 samples (4 metrics $\times$ 100).

3.3 Shortcut metrics

All shortcuts are deterministic and run locally:

- **ContextPrecision** $\rightarrow$ graded NDCG. A binary threshold variant collapses (about 89% of chunks are lexically relevant), so relevance is graded: $\sigma(\text{cross-encoder}(\text{reference}, \text{chunk}))$, and the metric is NDCG over graded relevance under the query-vs-chunk reranker order (retrieved and reranked variants; bge-reranker v2-m3 and ms-marco-MiniLM).
- **Faithfulness** $\rightarrow$ sentence-level NLI. Sentences of the response are split; each is entailed if $\max_c P_{\text{entail}}(c, s) \geq 0.5$ (DeBERTa-v3 MNLI); score = entailed / total.
- **AnswerRelevancy** $\rightarrow$ asymmetric QA embeddings. TAS-B dot product, and nomic cosine with `search_query:/search_document:` prefixes.
- **AnswerCorrectness** $\rightarrow$ score = $0.75 \text{ROUGE-L } F_1(\text{response}, \text{reference}) + 0.25 \cos(\mathbf{r}, \mathbf{ref})$ with **nomic-embed-text**.

3.4 Agreement analysis

For each metric pair we compute Pearson and Spearman correlation over the 99 joined samples. Verdict bands: $|\rho| \geq 0.7$ STRONG, $0.4 \leq |\rho| < 0.7$ MODERATE, $|\rho| < 0.4$ WEAK. Spearman is primary because TAS-B is on an unbounded scale.

4 Results

Table 1 summarizes the agreement. Full tables and figures are in the accompanying repository.

4.1 AnswerCorrectness is replaceable

The blend $0.75 \text{ROUGE-L} + 0.25 \cos$ reproduces the judge’s ranking almost perfectly ($\rho = 0.894$); each component alone is nearly as good (0.870, 0.875). For ranking and regression gating, any variant suffices; the judge adds value only where absolute calibration matters.

Table 1: Agreement between RAGAS judge metrics and deterministic shortcuts over 99 joined samples.

RAGAS metric	Shortcut	ρ (Spearman)	Verdict
AnswerCorrectness	combined	0.894	STRONG
AnswerCorrectness	ROUGE-L only	0.870	STRONG
AnswerCorrectness	nomic only	0.875	STRONG
AnswerRelevancy	nomic cosine	0.709	STRONG
AnswerRelevancy	TAS-B dot	0.576	MODERATE
Faithfulness	sentence NLI	-0.185	WEAK
ContextPrecision	bge (retrieved)	-0.124	WEAK
ContextPrecision	bge (reranked)	-0.044	WEAK
ContextPrecision	MiniLM	-0.018	WEAK

4.2 AnswerRelevancy: embedding family matters

The shortcut built from the retrieval index’s own embedding family (`nomic-embed-text`) with task prefixes agrees STRONGLY ($\rho = 0.709$). The fine-tuned TAS-B encoder is MODERATE ($\rho = 0.576$) and numerically saturated.

4.3 Faithfulness: the judge saturates

The judge’s median is 1.000 (everything is faithful) while the NLI shortcut’s median is 0.000 (nothing is entailed by the retrieved chunks). With a bimodal, near-degenerate judge, the correlation ($\rho = -0.185$) is driven by a few low-score outliers. Both measurements are plausible; neither can be trusted without a human oracle audit of 20–30 sentence-level entailment decisions.

4.4 ContextPrecision: no signal on this corpus

All variants are near-constant (bge NDCG mean ≈ 0.99 , std ≈ 0.01 ; RAGAS median 0.70). Every retrieved chunk is relevant on this corpus, so the metric cannot rank systems. This is a corpus degeneracy, not a general failure of the metric.

5 Discussion

Cheap wins for ranking. Two of the four judge metrics are reliably predicted by deterministic shortcuts that run in minutes rather than hours. This enables CI gating on the shortcut with periodic judge-based calibration.

Disagreement is a diagnostic. A near-zero correlation identifies either a judge artifact (faithfulness ceiling) or a corpus artifact (context precision degeneracy). Both findings are actionable and are surfaced at the cost of one short computation.

Threats to validity. A single judge and generator (both `gemma4:12b`) may inflate agreement relative to a frontier judge; the TAS-B scale limits inference to rank statistics; and the corpus has a ceiling on context relevance that may not generalize to other domains.

6 Conclusion

On this real agentic-RAG corpus, AnswerCorrectness and AnswerRelevancy can be evaluated cheaply and deterministically; Faithfulness is currently undecidable without an oracle audit; and ContextPrecision carries no discriminative power. The shortcut-versus-judge agreement check is cheap enough to run before adopting any RAG metric on a new corpus.

References

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